Gender Hiring Quotas

Valentina D'Anna* Eliana Sena†

(JOB MARKET PAPER)

Last update: November 1, 2026

Click here for the most updated version

Abstract

Gender quotas have mostly targeted corporate boards and done little for the gender employment gap. A quota on the stock of employment would force firms to separate incumbent workers or hire beyond their needs. A quota on new hires avoids these costs, but firms can comply and then undo its effect through separations. We study a unique policy: Italian firms that win public contracts funded by the EU recovery plan must reserve for women at least 30 percent of the new hires needed for the project. Coverage depends on how contracts are funded, so similar contracts are awarded with and without the quota. Linking procurement records to employer-employee data and balance sheets, we compare winners of contracts with and without the quota in a matched difference-in-differences design. The quota raises female hires by 26 percent, although some firms avoid it by hiring less. Two years after the award, female employment is 11.2 percent higher than in comparable winners without the quota. Male and total employment do not change significantly, so the gender composition of the workforce shifts toward women. Female separations fall while the quota is in force and, once contracts end, rise only in larger firms, so female employment remains above its pre-award level. The new jobs are mostly white-collar but temporary and part-time, with no effect on wages. We find no detectable costs for firms. Hiring quotas can bring women into male-dominated firms, and the gains outlast the obligation.

Keywords: Gender hiring quotas; Public procurement; Affirmative action; Labor demand; Female employment; Matched employer-employee data.
JEL Codes: H57, J16, J23, J63, J78.

*Rome Economics Doctorate (RED): EIEF, LUISS, Tor Vergata, and Sapienza.
†University of Pennsylvania.

1 Introduction

Despite decades of progress, gender gaps persist both in employment and in top positions (Olivetti and Petrongolo, 2016; Blau and Kahn, 2017). In OECD countries, 72.5 percent of women are employed, against 87.7 percent of men.[1] Quotas have become one of the most widespread tools to close these gaps. Since Norway's law in 2003, 28 countries have adopted binding quotas for corporate boards.[2] These quotas have raised women's presence on boards, but they reach a few, highly selected women. They can narrow the employment gap only indirectly, through spillovers to women below the top, and studies find little such effect (Bertrand et al., 2019; Maida and Weber, 2022). Whether a quota can close the gender gap in employment, and not only at the top, therefore remains an open question.

A quota on employment can regulate either the stock of workers or the flow of new hires. Regulating the stock is hard: a firm below the target would have to separate incumbent workers or hire beyond its optimal size, and forcing either can be legally controversial. A quota on new hires avoids both problems. It leaves existing matches intact and moves the firm toward the target as it hires new workers. Its effect on employment, however, is not obvious. Firms can comply and then undo the quota through separations, or place the new hires in marginal jobs. Whether a hiring quota works therefore depends on three questions. First, do more female hires (the flow) translate into more female employment (the stock)? Second, in which jobs do firms place the women they hire? Third, does the quota impose costs on firms?

We know little about the answers, for two reasons. First, governments find it hard to impose quotas on private employers, whose hiring they do not control. Second, where binding hiring targets exist, evaluating them requires data on every hire and separation in a firm, and a comparison group of similar firms not subject to the quota.[3] For these reasons, the evidence on gender quotas comes almost entirely from corporate boards and elected offices (Beaman et al., 2009; Baltrunaite et al., 2014; Bertrand et al., 2019; Maida and Weber, 2022).

We overcome these obstacles using a temporary gender hiring quota in Italy, which has

[1] Employment rates refer to the population aged 25-54 in 2024 (OECD, 2024).

[2] We count only binding quotas, with a numerical target and sanctions, and exclude soft rules such as comply-or-explain requirements, governance-code recommendations, and non-binding affirmative-action goals. After Norway: Iceland, Italy, France, Belgium, India, Germany, Portugal, Austria, South Korea, Greece, Argentina, the Netherlands, the United Arab Emirates, Malaysia, Ecuador, Denmark, Spain, Finland, Croatia, Lithuania, Malta, Slovakia, Slovenia, Estonia, Ireland, Romania, and the United States (California, 2018). Directive (EU) 2022/2381 extends board quotas to large listed firms across the EU.

[3] Existing affirmative-action programs in employment either set non-binding hiring goals (Leonard, 1984; Holzer and Neumark, 2000; Miller, 2017; Amano-Patino et al., 2021) or, when binding, target narrow groups of workers, such as persons with disabilities (Szerman, 2022).

one of the largest gender employment gaps in the EU.[4] Since 2021, firms that win public contracts funded by the National Recovery and Resilience Plan (PNRR)[5] must reserve for women at least 30 percent of the new hires needed for the project. The government thus imposes the quota on private firms using its purchasing power as a buyer rather than its authority as a regulator. Linking procurement records to employer-employee data, we compare winners of similar contracts with and without the quota. Our results deliver three main takeaways. First, the quota raises female hires by 26 percent, although some firms avoid it by hiring less. Two years after the award, female employment is 11.2 percent higher, with no change for men. Female separations fall while the quota is in force; once contracts end, they rise only in larger firms. The flow thus becomes stock: female employment stops growing once the quota expires, but remains above its pre-award level, shifting the gender composition of the firm's workforce. Second, the new jobs are mostly temporary and white-collar, and only 25 percent of the women hired come from other private firms.[6] Third, the quota imposes no detectable costs on firms.

Three features of the quota matter for our analysis.[7] First, firms know whether a contract carries the quota before deciding to bid, because contracting authorities specify it in the call for tenders.[8] Second, contracting authorities can waive the quota only with a formal justification, for instance because the contract is small or requires fewer than three hires. In practice, the quota applies in full to 30 percent of PNRR contracts, which account for 47 percent of their value, and is waived mostly in small contracts.[9] Third, after the award the quota binds only if the firm hires at least three workers for the project. It leaves the firm free to choose how many workers to hire and in which occupations, hours, and contract types. Firms can therefore comply by hiring more women or fewer men, or avoid the quota by staying below the three-hire threshold.

To our knowledge, this policy is unique: it is the first binding gender quota on the hiring

[4]In 2024, the employment rate of women aged 25-54 was 19.2 percentage points below that of men in Italy, twice the EU average of 9.6 points (Eurostat, 2024b).

[5]The PNRR is Italy's component of Next Generation EU, the EU's post-COVID-19 recovery program, running from 2021 to 2026. PNRR contracts accounted for 8.1 percent of the value of public procurement in 2021-2025, peaking at about 18 percent in 2023.

[6]The remaining 75 percent had no private-sector employee contract in the year before: they may have been out of the labor force, unemployed, self-employed, or employed in the public sector. So far, we cannot distinguish among these states in our data.

[7]The quota is set by Article 47 of Decree-Law 77/2021, which also sets an independent 30 percent quota on hires of workers under 36.

[8]Contracting authorities are the public entities that buy works, goods, and services, from ministries and regions to municipalities, health authorities, and publicly owned companies. They design the call for tenders, choose the award procedure, and select the winning firm.

[9]The quota is reduced in 2.1 percent of PNRR contracts and waived in 55.4 percent. The remaining 12.6 percent are unclassified. Section 2 describes the waiver rules.

of private firms that applies to all jobs and all levels of the workforce, while leaving firms full discretion over how to comply. It offers an ideal setting to answer our three questions. First, the quota binds. In the year before the award, the median firm in our matched sample employs 12 workers, one of whom is a woman, and hires three new workers, none of them women. Women account for only 9.8 percent of hires among all treated firms and 8.8 percent among matched treated firms, less than a third of the share the quota requires, and for 46 percent of firms complying would mean hiring a woman for the first time. Second, firms have strong incentives to comply, because the contract is worth on average 36.5 percent of their annual sales.[10] Third, the quota lasts only as long as the contract, so we can test whether its effects persist once it expires. The setting is also a demanding test. Nearly three quarters of the firms operate in construction, where only 8.5 percent of workers are women[11] and physical strength may give men a productivity advantage (Galor and Weil, 1996; Pitt et al., 2012). If the quota works here, it is likely to work where women are easier to hire.

We build a new dataset that links employer-employee records to the universe of Italian public procurement contracts and to firm balance sheets. Because coverage depends mainly on how a contract is funded, not on what it buys, we can compare firms awarded contracts with the quota to firms awarded comparable contracts without it, using a matched difference-in-differences design. Both groups win a public contract, so the comparison isolates the effect of the quota from that of winning. We match contracts on type of work, procedure, timing, and value, and firms on sector, size, and pre-award female employment share.[12] The key assumption is that, without the quota, two similar firms winning similar contracts would have hired and retained women in the same way.

An important concern is that firms know before bidding whether a contract carries the quota, so firms reluctant to hire women may avoid covered contracts. Three pieces of evidence suggest that selection is limited. First, we do not observe pre-trends in hiring or employment between treated and control firms. Second, we match firms on their pre-award female employment share. Third, covered contracts are large relative to the sales of the firms that win them, so a firm is unlikely to give one up to avoid hiring a woman. We cannot rule out selection entirely. We have recently obtained data on all bidders, which will allow us to study selection directly.

Relative to matched control firms, female hires rise by 26 percent of their baseline average

[10] Computed for the 88.5 percent of eligible treated firms for which we observe balance sheets. The median is 7.0 percent.

[11] In 2024, women accounted for 8.5 percent of workers aged 25-64 employed in construction in Italy (Eurostat, 2024a).

[12] The unit of analysis is the firm-contract, and treatment is a firm's first award of a contract with the quota. Before the award, treated and control firms employ on average 20.7 and 20.9 workers, and their contracts have nearly identical values.

in the two years after the award. Some firms avoid the requirement by hiring less: the probability of any hire falls by 2.8 percentage points, driven by fewer male hires. Our estimates capture the effect of exposure to the quota, an intention-to-treat effect, rather than the effect of compliance, which we do not observe directly. They also abstract from general equilibrium effects, such as changes in the hiring of control firms.

More female hires translate into more female employment. Two years after the award, female employment in treated firms exceeds that in control firms by 11.2 percent of its baseline average, about 0.3 additional women per firm, while male employment does not change. Total employment rises, but not significantly, so the composition of the workforce shifts toward women.

Separations determine how much of the flow becomes stock. While the quota is in force, firms retain more women: female separations fall in the award year. This runs against the concern that firms comply and then undo the quota by separating the women they hire. Once contracts end, female hiring returns to its pre-award level, and the response of separations depends on firm size. In levels, which give more weight to larger firms, female separations rise enough to match new female hires two years after the award. Scaled by baseline employment, separations do not rise. Larger firms thus undo part of the quota once it expires, while the average firm keeps the women it hired, and female employment remains stable above its pre-award level.

The new jobs are mostly temporary white-collar positions. Only 25 percent of the women hired come from other private firms; the rest had no private-sector employee contract in the previous year. Most are hired into white-collar occupations, but 81 percent of the new jobs are temporary and 64 percent part-time. We find no effect on wages.

The quota imposes no detectable costs on firms. Revenues, assets, and profit margins do not change, and labor productivity rises by 6.7 percent.

Two forces shape these responses. The first is the government's leverage as a buyer. Responses are stronger when the contract is large relative to the firm's wage bill, consistent with firms complying because they depend on public demand. The second is the combination of temporary contracts and hiring frictions, which explains why the gain stops growing but does not disappear. Temporary contracts can simply expire, which explains why larger firms separate some of the new hires once the quota ends. Yet replacing a woman with a man requires a new hiring cost, plus a firing cost for longer contracts, and she accumulates skills specific to the firm. The quota thus creates a wedge between keeping the new hires and returning to the old workforce composition, even after the quota expires. Even when the jobs end, they may strengthen women's attachment to the labor market: most of these women did not come from other private firms, and work experience can make it easier to

find the next job.

Contribution to the Literature Our paper contributes to three literatures. First, we add to the literature on gender quotas by moving the analysis from leadership positions to the workforce as a whole. Existing studies examine quotas for elected offices and corporate boards. These quotas can change beliefs about women in leadership (Beaman et al., 2009, 2012) and the quality of those selected (Baltrunaite et al., 2014). Their effects on firm performance are mixed: some studies find costs (Ahern and Dittmar, 2012; Matsa and Miller, 2013), others none (Eckbo et al., 2022; Ferrari et al., 2022). Because they move only a handful of positions, their effects on women's employment below the top are small (Bertrand et al., 2019; Maida and Weber, 2022). A hiring quota works through a different channel: it changes who enters the firm, at every level. We show that this channel raises female employment, mostly through temporary white-collar jobs, that part of the gain survives once the quota expires, and that firms bear no detectable costs, in contrast with the mixed evidence on board quotas.

Second, we contribute to the literature on affirmative action in employment. Existing programs fall into two groups. The first sets hiring goals for broad groups, as in the US federal contractor program, which ties affirmative action to public contracts but does not bind (Leonard, 1984; Holzer and Neumark, 2000; Kurtulus, 2016; Miller, 2017; Amano-Patino et al., 2021). The second binds, but applies to the stock of employment and to narrow groups, such as persons with disabilities (Szerman, 2022). We study a quota that is binding, applies to the flow of new hires, and targets half of the population. Because it expires with the contract, we can also measure how much of the response survives once the constraint is lifted. By following hires, separations, and the women who stay, we show how a flow quota turns into a stock.

Finally, we contribute to the literature on public procurement as a policy tool. Governments often use their purchasing power to favor particular suppliers, and existing work studies how these preferences affect who wins contracts, procurement costs, and competition (Marion, 2007, 2009; Krasnokutskaya and Seim, 2011; Athey et al., 2013; Rosa, 2024), and how winning a contract affects firm growth (Ferraz et al., 2015). Conditions on what winning firms do after the award have received less attention, and the main example, the US federal contractor program, sets non-binding goals. We show that a binding condition attached to public contracts can change the hiring of private employers at no detectable cost to firms, and that its effect grows with the firm's reliance on public demand.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting, Section 3 the data, and Section 4 the empirical strategy. Section 5 describes the

margins along which firms can respond to the quota and presents the results: the effects on female employment, the margins along which firms adjust, firm performance, and heterogeneity by the economic importance of the contract and by firm size. Section 6 concludes.

2 Institutional Setting

2.1 Public Procurement in Italy

Public procurement is one of the largest channels through which governments buy from private firms. Across OECD countries, it accounts for 12.7 percent of GDP (OECD, 2023). Italy is close to this average: public bodies spend more than €300 billion every year on works, services, and supplies.

The Italian procurement system is highly decentralized. Public calls are issued and managed by contracting authorities, including ministries, regions, municipalities, health agencies, and public companies. Each authority defines the contract object, chooses the award procedure, sets the reserve price, and selects the winner.[13]

Firms can bid alone or in a temporary consortium. The lead firm signs on behalf of all members, while all members remain jointly liable. Each contract binds the firm for a limited period, and all obligations attached to the contract apply over that period.

The contracting authority monitors execution and appoints a project manager and a works director or contract supervisor. A final inspection certifies performance before the last payment.[14]

Firms that do not comply face penalties and, in serious cases, termination. Serious breaches are recorded by ANAC and can exclude firms from future calls. For firms depending on public demand, non-compliance can therefore be costly beyond the single contract.

Winning a contract creates a labor demand shock. A firm can reallocate existing employees or hire new workers. This choice matters because the gender quota applies only to new hires.

2.2 The PNRR and the Gender Hiring Quota

The PNRR is Italy's share of NextGenerationEU. Between 2021 and 2026, it channels €191.5 billion into the Italian economy, largely through public procurement. After the common restrictions used to construct our procurement sample, PNRR funds finance 59,726 contracts

[13]Legislative Decree 50/2016, replaced by Legislative Decree 36/2023 from July 2023.

[14]The project manager is the *Responsabile unico del progetto*; the final inspection is the *collaudo* for works and the *verifica di conformità* for services and supplies.

worth €22.7 billion. Of these, 20,367 contracts worth €10.6 billion carry the full gender quota.

Article 47 of Decree-Law 77/2021 requires firms winning a covered contract to reserve for women at least 30 percent of the new hires needed to execute it.[15] A separate 30 percent requirement applies to workers younger than 36. We focus on the gender requirement.

The quota targets the flow of new hires, not the existing workforce. It does not prescribe occupations, wages, hours, or contract types. Like all contractual obligations, it applies only for the duration of the contract.

Contracting authorities can apply, reduce, or waive the quota, but deviations require an adequate and specific justification. Examples include small direct awards, projects expected to require fewer than three hires, specialized qualifications, and emergencies. A partial reduction is allowed where sectoral female employment is especially low. Across all PNRR contracts awarded in the study years, before these restrictions, 29.9 percent carry the full quota, 2.1 percent a reduced quota, 55.4 percent a waiver, and 12.6 percent remain unclassified.

The quota comes with possible penalties and reporting duties. Firms with 15 to 50 employees must report the gender composition of their workforce within six months. Failure to report excludes them from subsequent PNRR tenders for twelve months. Enforcement is nevertheless uncertain. We observe exposure to the quota, not compliance, and estimate the policy's effect as monitored and enforced in practice.

2.3 A Hiring Quota for the Entire Workforce

The Italian quota differs from board and political quotas because it acts on the hiring of ordinary workers (Bertrand et al., 2019; Maida and Weber, 2022; Beaman et al., 2009; Baltrunaite et al., 2014). It also differs from non-binding affirmative-action plans and stock-based disability quotas (Leonard, 1984; Miller, 2017; Szerman, 2022). It sets a binding numerical target on the flow of hires and expires with the contract.

Three features make the policy well suited to study firm responses. It binds when firms expand; quota status varies across contracts with similar economic content; and the obligation is temporary. Italy is also a setting in which the quota may matter greatly. Among eligible treated firms, women account for 10.9 percent of employees and 9.8 percent of hires, far below the 30 percent requirement.

[15] Decree-Law 77/2021, converted into Law 108/2021. Implementing guidelines were adopted on December 7, 2021.

2.4 Where the Quota Binds

The quota falls mainly on sectors with few women. Figure 1 reports sector shares among the 4,112 treated firm-contracts and the female employment distribution of winning firms.

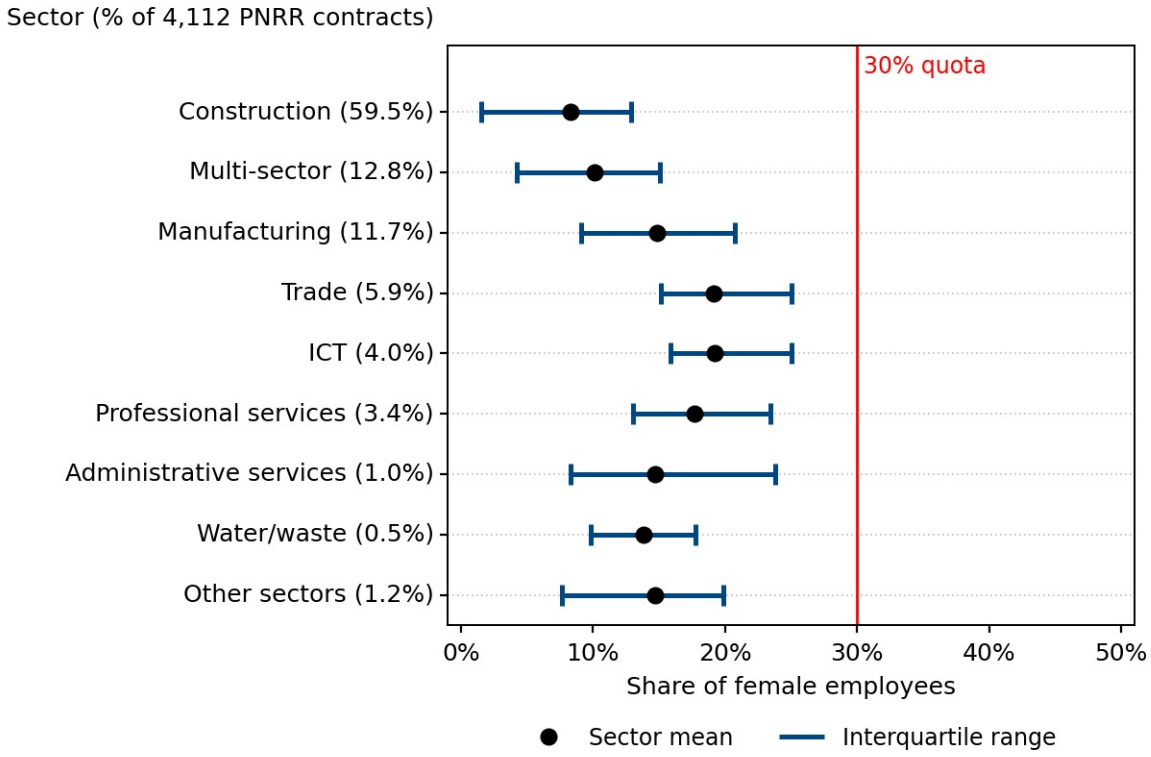


Figure 1: Exposure to the Quota by Sector

Notes: The figure reports each sector's share of the 4,112 treated firm-contracts subject to the full quota and the female employment distribution among winners.

The typical winning firm employs far fewer women than the quota requires. Moreover, 68.8 percent of eligible treated firms hired at least three workers in the baseline year. Among matched treated firms with positive baseline hiring, only 9.5 percent already hired at least 30 percent women. Most firms would therefore need to alter hiring composition when the threshold is reached.

Construction firms make up 70.9 percent of all treated firms and 73.4 percent of matched treated firms. Construction is therefore a demanding test: firms may avoid hiring or place women in a narrow set of jobs.

3 Data: Linking Contracts to Workers

Our data link Italian public contracts awarded from 2017 to 2025 to the full workforce of each winning firm. We combine ANAC procurement records, INPS matched employer-employee data, ReGiS quota information, and Orbis balance sheets.

3.1 Firm-Level Administrative Data

INPS records contain every private-sector employment contract from 2017 to 2025. We observe wages, weeks worked, working time, contract type, occupation, age, and gender. We construct employment, hires, separations, and their gender composition at the firm-year level. A hire is a worker employed by the firm in a year but not in the previous year. A hire has no prior private-sector contract if the worker held no private employment contract in the previous year; this group may include unemployed, inactive, self-employed, or public-sector workers.

Orbis provides sales, input costs, value added, profits, and total assets. We measure labor productivity as log value added per employee.

3.2 Public Procurement Data

ANAC records provide contract object, CPV code, procedure, location, amount, authority, funding source, and winners. ReGiS provides quota status for PNRR calls issued from 2021 to 2024. We link sources through the tender identifier.

3.3 Sample Restrictions

We focus on contracts awarded between 2021 and 2024 and worth between €15,000 and €10 million. We keep private firms with at least one baseline employee and retain only the lead firm in a consortium. The unit is the firm-contract. The final sample contains 214,073 firm-contracts won by 45,593 firms.

3.4 Descriptive Statistics

PNRR contracts represent 6.8 percent of contracts in our procurement sample and 7.3 percent of their total value. The median PNRR contract is worth €61,537, compared with €72,484 for an ordinary contract. Works account for 34.8 percent of PNRR contracts and 19.7 percent of ordinary contracts.

PNRR contracts are slightly shorter among completed contracts with observed duration. Their median duration is 4.6 months, compared with 5.5 months for ordinary contracts. Moreover, 84.5 percent last less than one year, compared with 78.9 percent of ordinary contracts. Duration is observed for 20.6 percent of PNRR contracts and 16.8 percent of ordinary contracts. Because it is observed only for completed contracts, observed duration understates true duration.

Table 1: Public Contracts and the Gender Hiring Quota

	Ordinary	PNRR
Number of contracts	823,096	59,726
Share of contracts (%)	93.2	6.8
Total value (€ billion)	287.4	22.7
Share of total value (%)	92.7	7.3
Median value (€)	72,484	61,537
Works (%)	19.7	34.8
Duration observed (%)	16.8	20.6
Median observed duration (months)	5.5	4.6
Observed duration below one year (%)	78.9	84.5

Notes: The table uses the matching-eligible procurement universe. Duration is observed only for completed contracts.

Waivers are more common than the full quota, but the full quota covers greater value. It applies to 29.9 percent of PNRR contracts and 46.7 percent of PNRR value. Waivers apply to 55.4 percent of contracts and 18.0 percent of value. In the matching-eligible sample, the median full-quota contract is worth €95,000, compared with €47,780 for waived contracts. The most common coded waiver reason is a reduced contract amount, reported for 14.1 percent of waived contracts.

The eligible treated sample contains 4,112 firms, while the potential-control pool contains 209,961 firm-contracts won by 41,481 firms. The median eligible treated firm employs 18 workers, compared with three in the private sector. Women account for 10.9 percent of employment and 9.8 percent of hires, compared with 40.1 percent in the private sector.

Hiring is common among eligible treated firms. In the baseline year, 91.8 percent hire at least one worker and 68.8 percent hire at least three. The median firm hires five workers but no women. Female hires are less often temporary than male hires, but more often part-time and white-collar. In the matched sample, 50.8 percent of baseline female hires had no prior private-sector contract, and only 9.5 percent of firms with positive hiring already hire at least 30 percent women.

Eligible treated firms are frequent procurement participants. Between 2017 and 2020, the median treated firm won six public contracts, and 39.5 percent won at least one contract

Table 2: Baseline Characteristics of Firms Exposed to the Full Quota

	Eligible treated	Matched treated	Private sector
Firm-contracts	4,112	685	—
Median employment	18	12	3
Median hires	5	3	—
Median female hires	0	0	—
Any hire (%)	91.8	87.4	—
At least three hires (%)	68.8	55.0	—
Female employment share (%)	10.9	11.1	40.1
Female share of hires (%)	9.8	8.8	40.1
Female hiring share at least 30% (%)	—	9.5	—
Female temporary hires (%)	35.7	41.6	—
Male temporary hires (%)	58.1	59.2	—
Female part-time hires (%)	21.8	33.9	—
Male part-time hires (%)	4.9	5.6	—
Female white-collar hires (%)	89.4	88.9	—
Male white-collar hires (%)	24.1	19.1	—
Female hires with no prior private-sector contract (%)	—	50.8	—

Notes: Characteristics are measured in the baseline year. The 30 percent benchmark is computed among matched treated firms with positive baseline hiring.

each year. Among firms appearing in the full-quota and waiver winner pools, 18.9 percent win both types.

Quota contracts are economically important. Among eligible treated firms with balance-sheet data, the first quota contract is worth on average 36.5 percent of annual sales and 7.0 percent at the median. The median contract equals 0.62 times annual wage bill. For 25 percent of firms, it exceeds 28.7 percent of sales, and for 10 percent it exceeds 84.4 percent. Sales are observed for 88.5 percent of eligible treated firms.

4 Empirical Strategy

4.1 Comparing Winners of Similar Contracts

The main empirical challenge is selection into contracts with the quota. Contracting authorities decide quota status before publication, and firms observe it before bidding. A simple comparison would therefore mix the quota's effect with differences in firms, sectors, authorities, and projects.

We compare firms winning nearly identical contracts, with and without the quota. Every firm in the comparison wins a public contract and therefore faces a similar demand shock. The central difference is that women must make up at least 30 percent of any new hires

needed by the treated firm.

4.2 Treatment and Comparison Groups

Treatment is defined at the firm level. For each firm, we keep its first PNRR contract subject to the full quota. This leaves 4,112 treated firm-contracts.

Control firm-contracts carry no quota. They include ordinary contracts and PNRR contracts with a complete waiver. We exclude contracts with a reduced quota. Firms that ever win a full-quota contract cannot serve as controls. The potential-control pool contains 209,961 firm-contracts won by 41,481 firms.

The same control firm can appear more than once because it may win several eligible contracts and because matching is with replacement. We account for this repetition in the standard errors.

4.3 Matching on Contracts and Firms

We match each treated firm-contract to one control firm-contract in two steps. First, contracts must have the same object and award procedure, be awarded within six months, and share a CPV code. We first search at four digits, then three, and then two.[16] Contract values must differ by less than 10 log points and €1,500.

Second, firms must operate in the same macro-sector. Log baseline employment must differ by less than 0.30, and the female employment share by less than 10 percentage points. Among candidates, we select the nearest control using standardized Euclidean distance.

The procedure yields 685 matched pairs, or 16.7 percent of treated firm-contracts. The 685 controls correspond to 665 distinct firm-contracts won by 642 firms. Table 3 shows that treated and control firms are nearly identical before the award. The average firm employs about 21 workers, of whom 2.6 are women, and wins a contract worth about €92,000.

[16] In the matched sample, 73 percent of pairs share a four-digit CPV code, 13 percent a three-digit code, and 14 percent a two-digit code.

Table 3: Baseline balance between all treated firms, matched treated firms, and matched controls

	All treated		Matched treated		Matched controls	
Variable	Mean	Median	Mean	Median	Mean	Median
Panel A: Variables used in matching						
Female employment share	0.11	0.10	0.11	0.11	0.11	0.11
Total employment (log)	3.10	2.89	2.63	2.48	2.63	2.48
Industry	0.13		0.13		0.13	
Services	0.16		0.13		0.13	
Construction	0.59		0.70		0.70	
Agriculture	0.00		0.00		0.00	
Multisector	0.12		0.04		0.04	
Procurement amount	836,793.47	140,667.52	92,790.81	47,638.07	92,799.21	47,607.00
Panel B: Main baseline outcomes						
Total employment	43.63	18.00	21.39	12.00	21.51	12.00
Female employment	5.39	2.00	2.63	1.00	2.60	1.00
Male employment	38.24	17.00	18.75	11.00	18.91	11.00
Blue-collar employment	25.17	12.00	13.73	9.00	14.32	9.00
White-collar employment	18.46	4.00	7.66	3.00	7.19	2.00
Weeks worked	1,684.72	653.00	839.24	463.00	834.05	468.00
Wage bill	1,207,789.97	367,138.50	556,132.69	251,897.00	526,392.38	252,418.00
White-collar employment share	0.32	0.22	0.27	0.20	0.25	0.18
Female white-collar employment share	0.10	0.09	0.11	0.10	0.10	0.10
Total hires	11.09	5.00	4.95	3.00	5.12	3.00
Female hires	1.08	0.00	0.44	0.00	0.45	0.00
Average weekly wage	576.79	524.18	545.12	508.44	546.84	510.56
Average weekly wage female	598.14	543.95	561.14	528.10	564.49	524.83
Average weekly wage (log)	6.25	6.23	6.21	6.20	6.22	6.20
Gender wage gap (%)	-1.44	-0.59	-1.94	-1.74	-2.79	-1.98
N. observations	4,112		685		685	
N. firm-events	4,112		685		665	
N. unique firms	4,112		685		642	

Notes: Statistics are measured in the baseline year. "All treated" reports eligible treated firm-events before matching; "Matched treated" and "Matched controls" report the matched firm-events used in the estimation. A control firm matched to different procurement events enters once for each event, possibly with different baseline years. Panel A reports the firm-level matching variables and Panel B selected baseline outcomes. The female share of white-collar employment is computed for firms with at least one white-collar employee, and the average female weekly wage for firms with at least one female employee. The gender wage gap is the log difference in firm-level FTE weekly wages between men and women, read as the percentage by which men's wages exceed women's, and is computed for firms with at least one worker of each gender.

The matched sample differs from all treated firms. Matched firms have median employment of 12 workers, compared with 18 among all treated firms. Their median contract is about €48,000, compared with €141,000. Construction accounts for 73.4 percent of matched treated firms and 70.9 percent of all treated firms. Baseline female employment shares are approximately 11 percent in both groups.

4.4 Estimating Equation

We estimate a difference-in-differences model within matched pairs. For firm i, contract c, pair p, and year t:

$$Y_{icpt} = \beta \left(\text{Treated}_{icp} \times \text{Post}_{ict}\right) + \alpha_{ic} + \lambda_{pt} + \varepsilon_{icpt}. \tag{1}$$

Firm-by-contract fixed effects, α_{ic}, absorb time-invariant differences. Pair-by-year fixed effects, λ_{pt}, absorb shocks common to each pair. The coefficient β measures the average quota effect on treated firms in the matched sample. We cluster standard errors at the firm level.

We also estimate:

$$Y_{icpt} = \sum_{k \neq -1} \beta_k \left(\text{Treated}_{icp} \times \mathbf{1}[t - E_{ic} = k]\right) + \alpha_{ic} + \lambda_{pt} + \varepsilon_{icpt}, \tag{2}$$

where the year before the award is the reference period. Pre-award coefficients test for differential trends.

4.5 Identifying Assumption and Threats

The identifying assumption is that treated firms and matched controls would have followed parallel paths without the quota. Matching makes this assumption more plausible, and event studies show no evidence of differential pre-trends.

Three concerns remain. Firms may select into quota calls on unobserved characteristics. Treated and control firms may compete in the same labor markets. Finally, we observe exposure rather than compliance, so our estimates identify the effect of the quota as implemented in practice. The concurrent youth quota is a further feature to keep in mind when reading the results: covered firms must also reserve 30 percent of new hires for workers under 36. Further analysis of the characteristics of the women hired is needed to assess how the two requirements interact.

5 How Firms Respond to a Hiring Quota

The quota can affect whether firms hire, how many women and men they hire, how long the new hires stay, and the jobs offered to women. These margins are not mutually exclusive. All estimates are relative to matched controls, so they isolate the effect of the quota from the common effect of winning a contract. They capture the effect of exposure to the quota, an intention-to-treat effect, rather than the effect of compliance, which we do not observe directly. They also abstract from general equilibrium effects, such as changes in the hiring of control firms.

5.1 First Stage: Female Hiring

The quota raises female hiring (Figure 2). Treated firms hire 0.11 more women per year than controls, an increase of 25.9 percent over a baseline of 0.44 (panel a). Scaled by baseline employment, the female hiring rate rises by 0.7 percentage points, or 35.8 percent of its baseline value of 2.0 percent (panel b). The increase is concentrated in the award year and the following year.

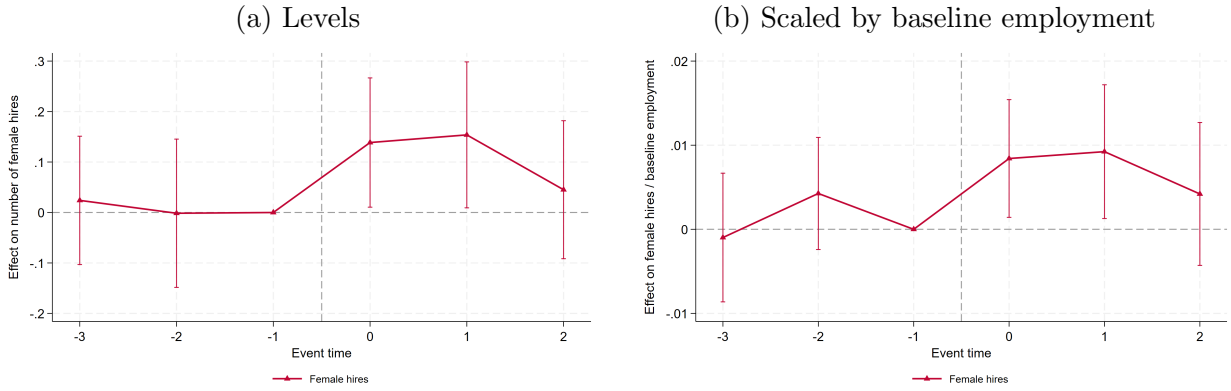


Figure 2: Female Hires: Levels and Scaled

Notes: Event-study coefficients compare treated firms with matched controls. Event time zero is the award year; the preceding year is omitted. Bars show 95 percent confidence intervals with standard errors clustered at firm level. Panel (a) reports effects on the number of female hires. Panel (b) reports effects on female hires divided by baseline employment.

Some firms avoid the quota by hiring less. A firm must comply only if it hires at least three workers for the project, so it can avoid the constraint by reallocating existing workers or staying below the threshold. The probability of any hire falls by 2.8 percentage points from a baseline of 87 percent. The probability of hiring a man falls by 3.0 percentage points, while the probability of hiring a woman does not change.

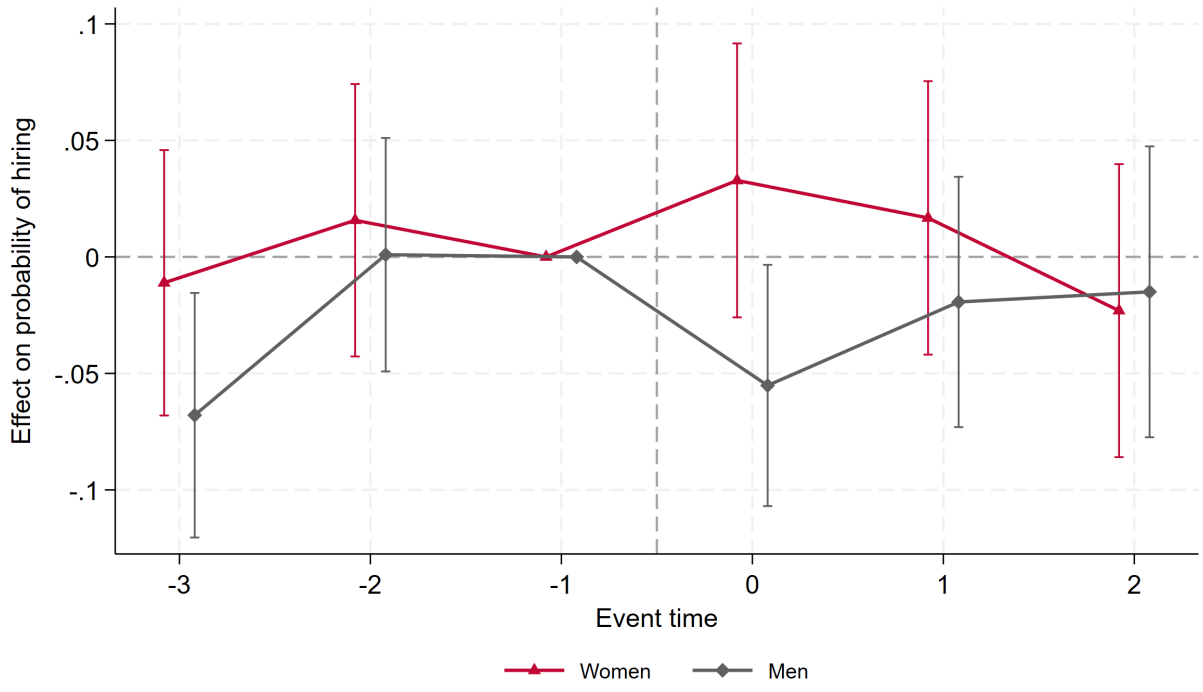


Figure 3: Probability of Hiring, by Gender

Notes: Event-study coefficients compare treated firms with matched controls. The outcome is an indicator for hiring at least one worker of each gender.

5.2 Female Employment

More female hires translate into more female employment. Two years after the award, female employment in treated firms exceeds that in control firms by 11.2 percent of its baseline average, about 0.3 additional women per firm over a baseline of 2.6. Male employment does not change. Total employment rises, but not significantly, so the gender composition of the workforce shifts toward women. Before the award, treated and control firms follow similar paths.

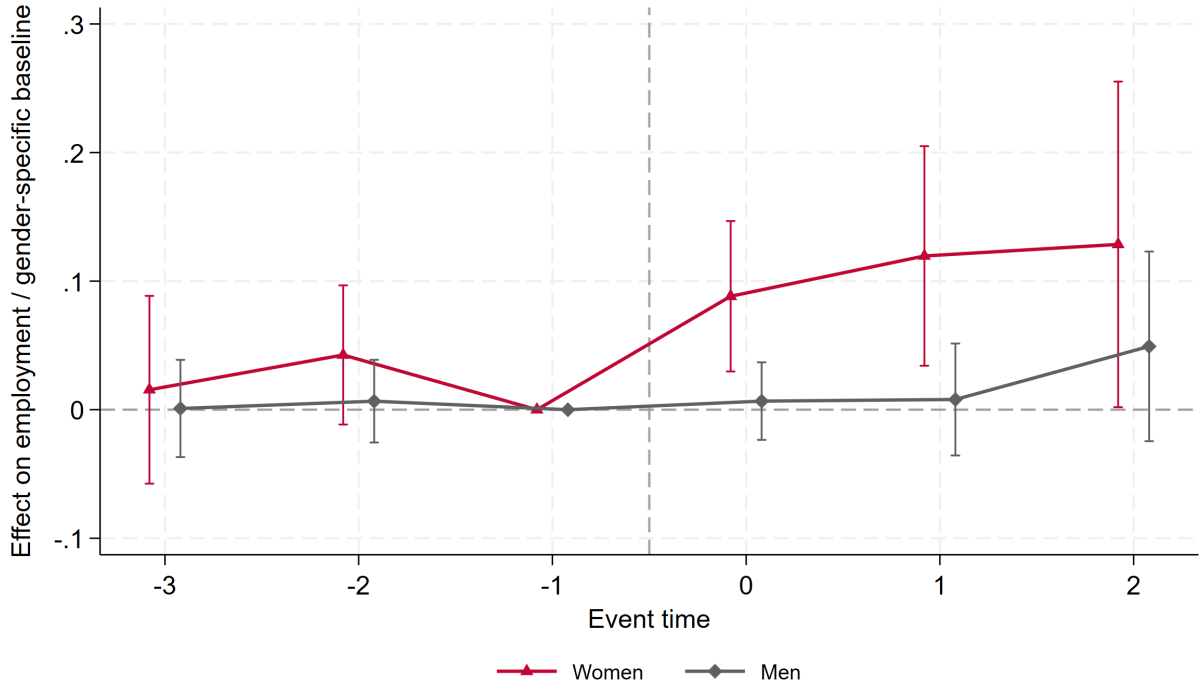


Figure 4: Employment by Gender

Notes: Event-study coefficients compare treated firms with matched controls. Event time zero is the award year; the preceding year is omitted. Bars show 95 percent confidence intervals with standard errors clustered at firm level. Outcomes are female and male employment scaled by their baseline averages.

5.3 From Flow to Stock: Separations

Separations determine how much of the flow becomes stock. While the quota is in force, firms retain more women: female separations fall in the award year, both in levels and scaled by baseline employment (Figure 5). This runs against the concern that firms comply and then undo the quota by separating the women they hire.

Once contracts end, female hiring returns to its pre-award level, and the response of

separations depends on firm size. In levels, which give more weight to larger firms, female separations rise enough to match new female hires two years after the award (panel a). Scaled by baseline employment, separations do not rise (panel b). Larger firms thus undo part of the quota once it expires, while the average firm keeps the women it hired, and female employment remains stable above its pre-award level. The rise in separations comes two years after the award, although observed contract durations are short. Because duration is observed only for completed contracts, it understates true duration (Section 3.4), and many contracts may still be active in the year after the award.

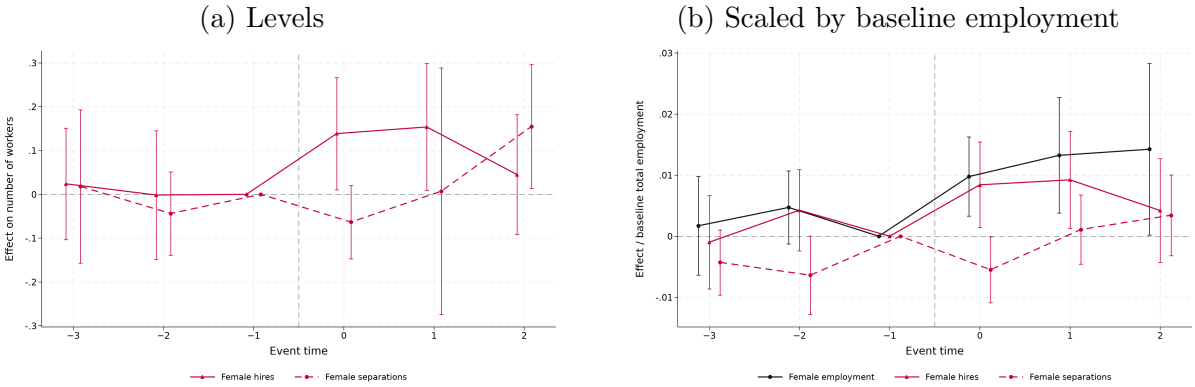


Figure 5: Female Hires and Separations: Levels and Scaled

Notes: Event-study coefficients compare treated firms with matched controls. Event time zero is the award year; the preceding year is omitted. Bars show 95 percent confidence intervals with standard errors clustered at firm level. Panel (a) reports effects on the number of female hires and separations. Panel (b) reports effects on female employment, hires, and separations divided by baseline total employment. Separations are women employed in the previous year who are no longer employed by the firm.

5.4 Which Women, in Which Jobs, and on What Terms

Only 25 percent of the additional women come from other private firms. Hires with no prior private-sector contract account for 26.8 percentage points of the 35.8 percent increase in the female hiring rate, and hires from other private firms for 9.0 points. The remaining 75 percent may have been out of the labor force, unemployed, self-employed, or employed in the public sector, which we cannot yet distinguish. Before the award, 50.8 percent of female hires in matched treated firms had no prior private-sector contract, so the quota tilts hiring further toward women outside private employment.

The new jobs are mostly temporary white-collar positions. White-collar hires account for 26.3 percentage points of the increase and blue-collar hires for 9.5 points. Temporary contracts account for 81 percent of additional female hires and part-time contracts for 64

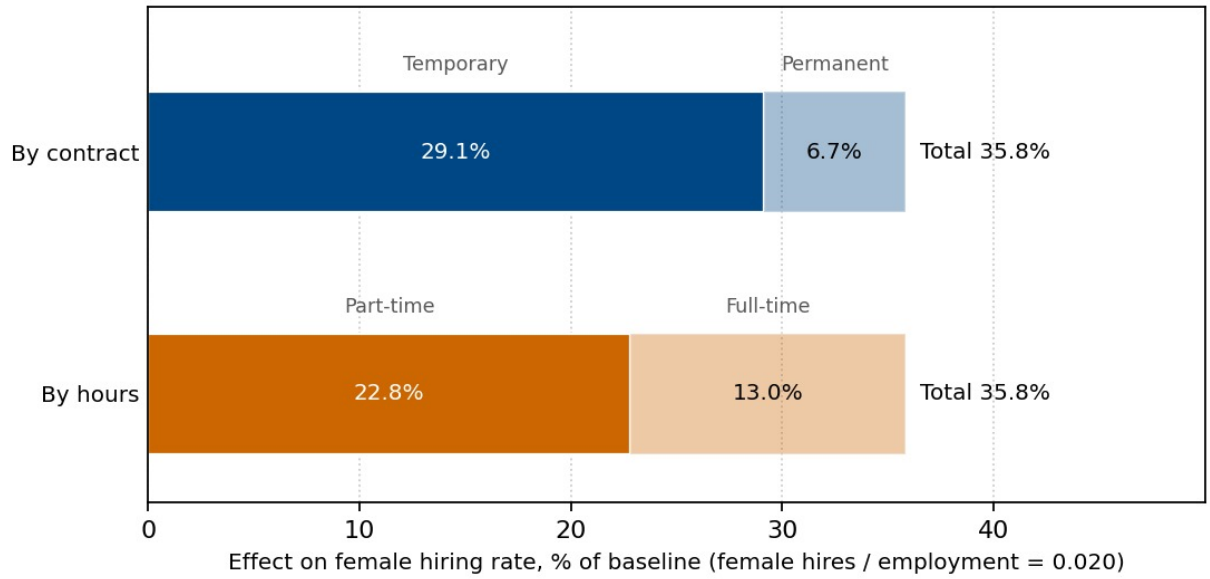


Figure 6: Composition of Additional Female Hires

Notes: The figure decomposes the post-award increase in the female hiring rate by previous labor-market status and occupation.

percent, well above the baseline shares of 41.6 and 33.9 percent among female hires in matched treated firms.

Weekly wages do not change. Estimates for women and men, new hires, and white-collar workers are close to zero.

5.5 Costs on Firms

The quota imposes no detectable costs on firms. Revenues, total assets, and profit margins do not change, and labor productivity, measured as value added per employee, rises by 6.7 percent. Because many additional women work part-time, a per-employee measure, if anything, understates productivity per hour worked.

5.6 Heterogeneity

Responses are concentrated where the contract matters most. We measure importance as contract value relative to the treated firm's average annual wage bill in the two years before the award. In the top tercile, the probability of any hire falls and the female employment share rises. Effects in the bottom tercile are not distinguishable from zero.

Responses also differ by firm size. Small firms drive the employment gains: female weeks worked and female hiring rise sharply. Mid-sized firms reduce hiring. Larger firms account

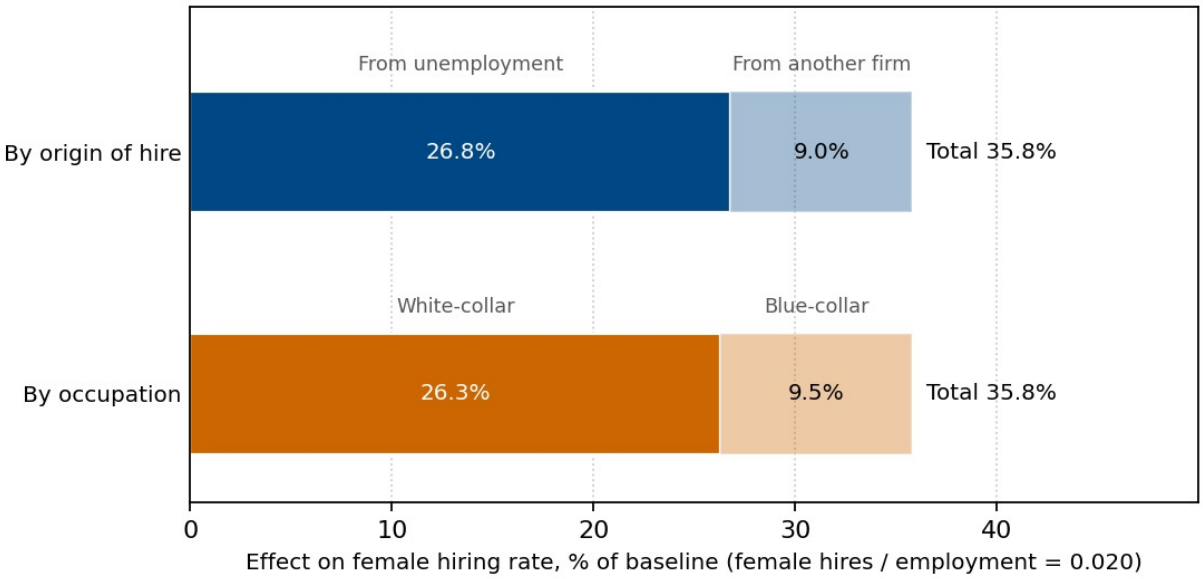


Figure 7: Terms of Employment of Additional Female Hires

Notes: The figure decomposes the increase in the female hiring rate by contract type and working time.

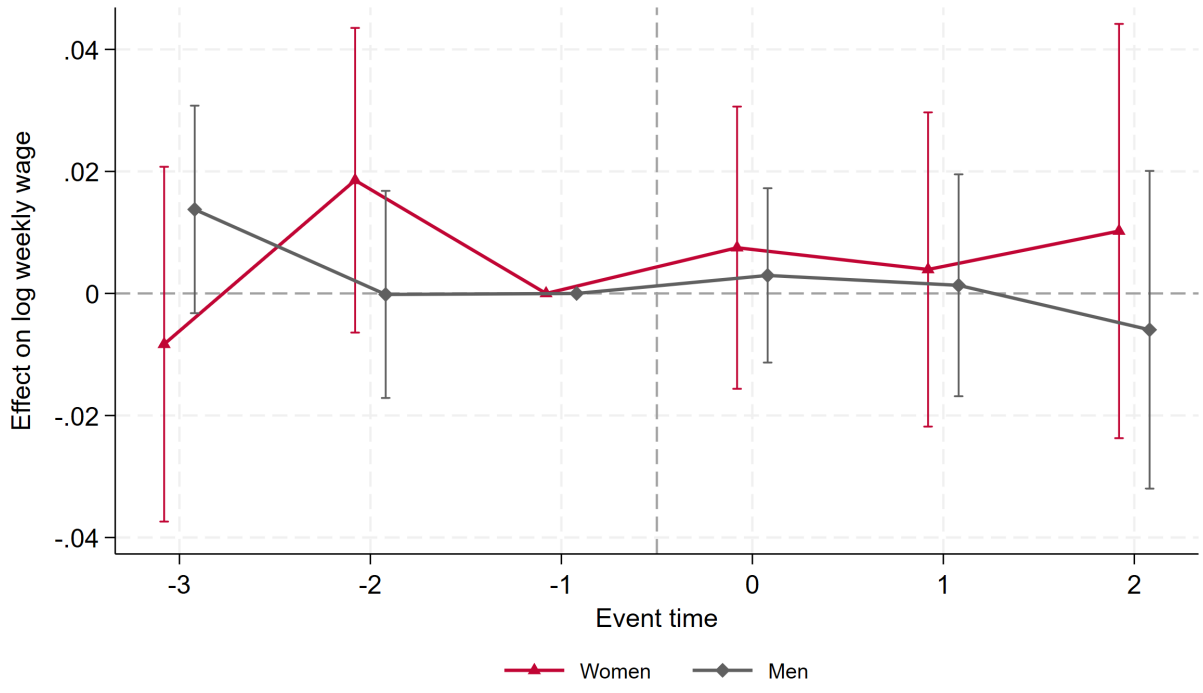


Figure 8: Weekly Wages by Gender

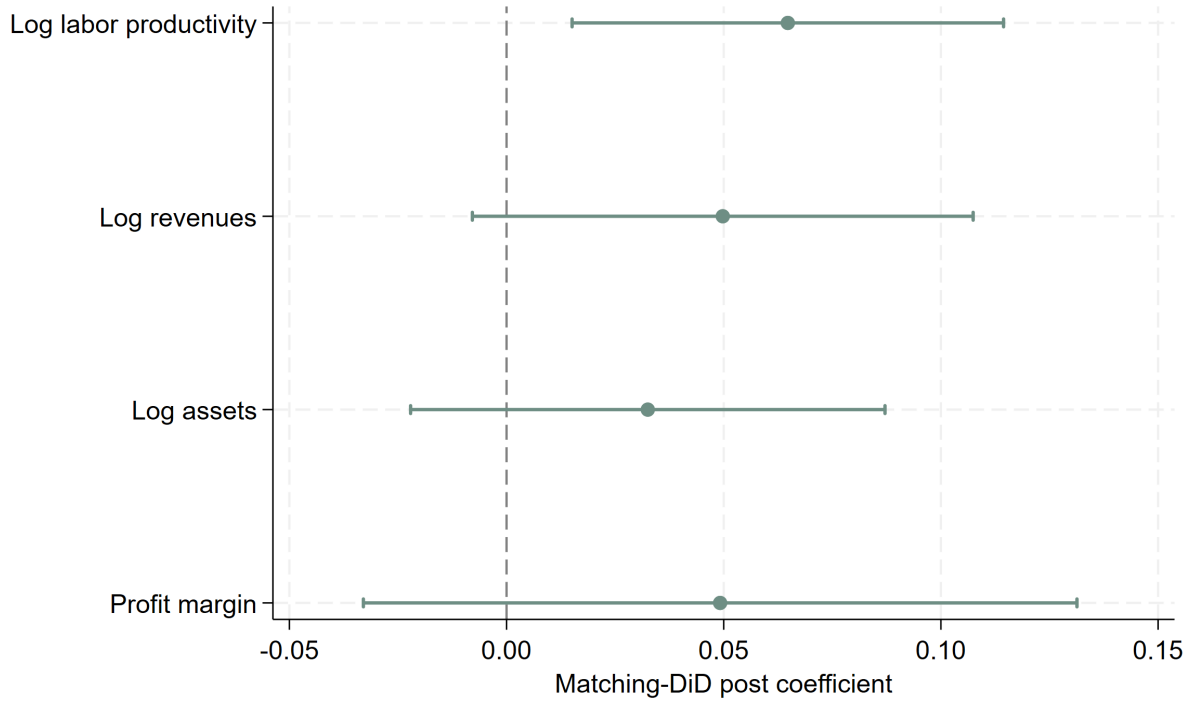


Figure 9: Firm Performance

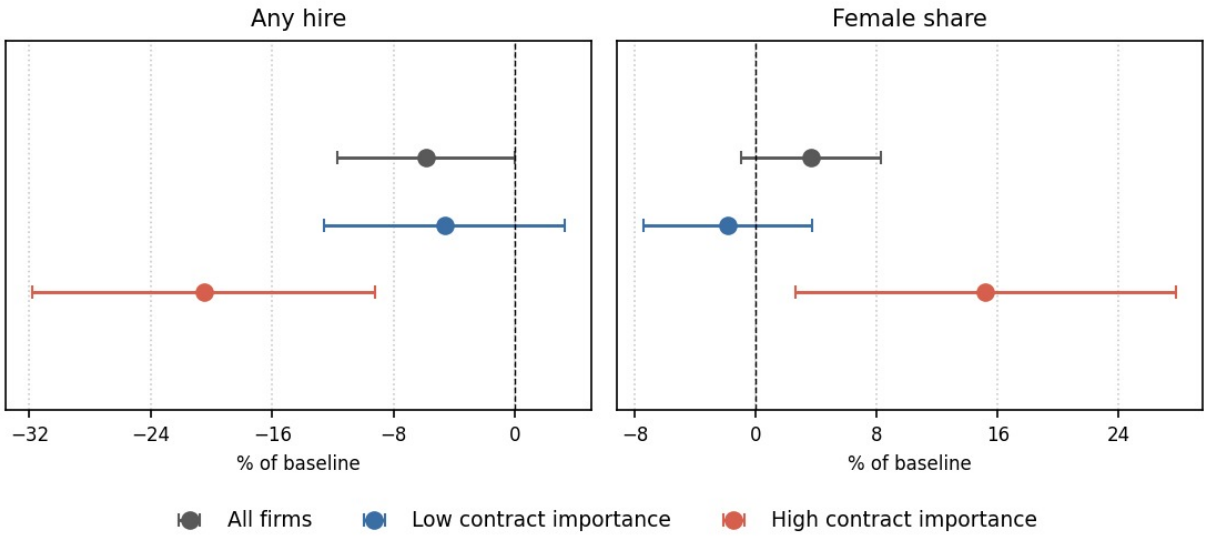


Figure 10: Heterogeneity by Contract Importance

for the rise in female separations after contracts end, which is why separations rise in levels but not when scaled by baseline employment.

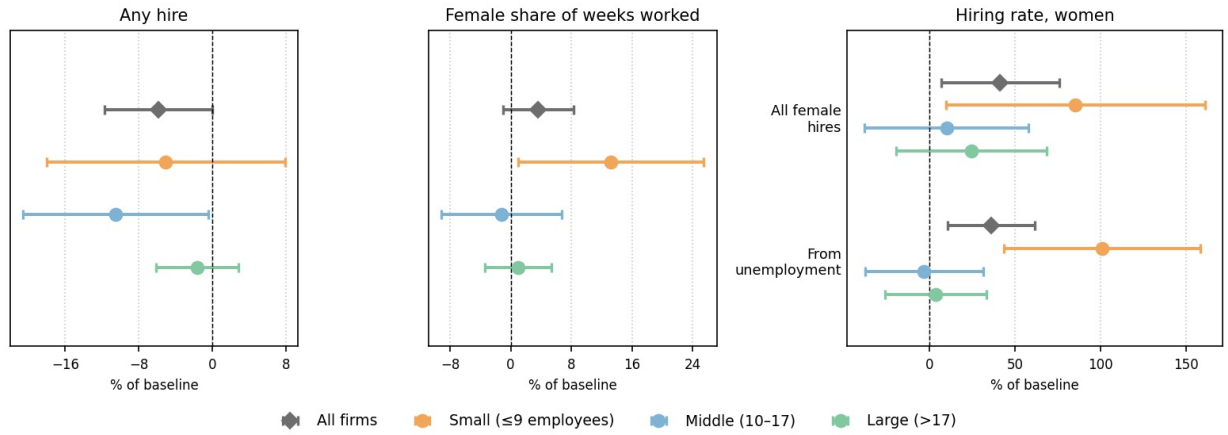


Figure 11: Heterogeneity by Firm Size

6 Conclusions

We study a gender quota on new hires attached to public contracts in Italy. Firms exposed to the quota hire more women than comparable firms winning similar contracts without it, and two years after the award they employ 11.2 percent more women, with no significant change in male or total employment. Some firms avoid the requirement by hiring less. Female separations fall while the quota is in force and, once contracts end, rise only in larger firms, so the gender composition of the workforce remains shifted toward women. Only 25 percent of the additional women come from other private firms, and most enter white-collar jobs that are temporary and part-time. We find no detectable costs for firms.

Three lessons follow. First, binding conditions attached to public contracts can change the hiring of private employers. Governments rarely regulate whom private firms hire, but as buyers they can attach conditions to the contracts they award, and firms accept them when the contract matters. The reach of this lever grows with firms' reliance on public demand. Second, a quota on the flow of hires can change the stock of employment. Firms did not undo the quota while it was in force, and most kept the women they hired after it expired. Temporary contracts, however, allowed larger firms to undo part of the gain. A hiring quota is therefore more likely to have lasting effects when the jobs it creates are not designed to end with the obligation. Third, even short jobs may strengthen women's attachment to the labor market. Most of the additional women did not come from other private firms, so the

quota may have drawn women into private employment rather than moving them across firms.

Our results come with caveats. We estimate the effect of exposure to the quota as enforced in practice, not the effect of full compliance, and we abstract from general equilibrium effects. The treatment combines the quota with a large public contract, so the effects may differ for firms that face a quota without a demand shock. The matched sample consists mostly of small construction firms, where women are hard to hire. This makes the setting a demanding test, but the magnitudes may not carry over to sectors with a larger supply of female workers. Finally, firms know before bidding whether a contract carries the quota, and we cannot yet rule out selection entirely.

Several questions remain open. Data on all bidders will allow us to study selection into covered contracts directly. Linking the women hired under the quota to their later careers will show whether these jobs lead to stable employment, and data on public employment will separate women drawn from outside the labor force from those moving out of the public sector. A closer look at contract duration will clarify how long the obligation binds and why separations rise when they do. Finally, scaling our estimates to the Italian economy would show how much a hiring quota of this kind could narrow the gender employment gap.

References

Ahern, K. R. and Dittmar, A. K. (2012). The changing of the boards: The impact on firm valuation of mandated female board representation. *The quarterly journal of economics*, 127(1):137–197.

Amano-Patino, N., Aramburu, J., and Contractor, Z. (2021). The effect of affirmative action on workers' outcomes.

Athey, S., Coey, D., and Levin, J. (2013). Set-asides and subsidies in auctions. *American economic journal: microeconomics*, 5(1):1–27.

Baltrunaite, A., Bello, P., Casarico, A., and Profeta, P. (2014). Gender quotas and the quality of politicians. *Journal of Public Economics*, 118:62–74.

Beaman, L., Chattopadhyay, R., Duflo, E., Pande, R., and Topalova, P. (2009). Powerful women: Does exposure reduce bias? *Quarterly Journal of Economics*, 124(4):1497–1540.

Beaman, L., Duflo, E., Pande, R., and Topalova, P. (2012). Female leadership raises aspirations and educational attainment for girls: A policy experiment in india. *Science*, 335(6068):582–586.

Bertrand, M., Black, S. E., Jensen, S., and Lleras-Muney, A. (2019). Breaking the glass ceiling? the effect of board quotas on female labour market outcomes in norway. *Review of Economic Studies*, 86(1):191–239.

Blau, F. D. and Kahn, L. M. (2017). The gender wage gap: Extent, trends, and explanations. *Journal of Economic Literature*, 55(3):789–865.

Eckbo, B. E., Nygaard, K., and Thorburn, K. S. (2022). Valuation effects of norway's board gender-quota law revisited. *Management Science*, 68(6):4112–4134.

Eurostat (2024a). Employment by sex, age and economic activity (NACE rev. 2) (lfsa_egan2). Eurostat database, Labour Force Survey.

Eurostat (2024b). Employment rates by sex, age and citizenship (lfsa_ergan). Eurostat database, Labour Force Survey.

Ferrari, G., Ferraro, V., Profeta, P., and Pronzato, C. (2022). Do board gender quotas matter? selection, performance, and stock market effects. *Management Science*, 68(8):5618–5643.

Ferraz, C., Finan, F., and Szerman, D. (2015). Procuring firm growth: The effects of government purchases on firm dynamics. NBER Working Paper 21219, National Bureau of Economic Research.

Galor, O. and Weil, D. N. (1996). The gender gap, fertility, and growth. *The American Economic Review*, 86(3):374–387.

Holzer, H. and Neumark, D. (2000). Assessing affirmative action. *Journal of Economic literature*, 38(3):483–568.

Krasnokutskaya, E. and Seim, K. (2011). Bid preference programs and participation in highway procurement auctions. *American Economic Review*, 101(6):2653–2686.

Kurtulus, F. A. (2016). The impact of affirmative action on the employment of minorities and women: A longitudinal analysis using three decades of EEO-1 filings. *Journal of Policy Analysis and Management*, 35(1):34–66.

Leonard, J. S. (1984). The impact of affirmative action on employment. *Journal of Labor Economics*, 2(4):439–463.

Maida, A. and Weber, A. (2022). Female leadership and gender gap within firms: Evidence from an italian board reform. *ILR Review*, 75(2):488–515.

Marion, J. (2007). Are bid preferences benign? the effect of small business subsidies in highway procurement auctions. *Journal of Public Economics*, 91(7-8):1591–1624.

Marion, J. (2009). How costly is affirmative action? government contracting and california's proposition 209. *The Review of Economics and Statistics*, 91(3):503–522.

Matsa, D. A. and Miller, A. R. (2013). A female style in corporate leadership? evidence from quotas. *American Economic Journal: Applied Economics*, 5(3):136–169.

Miller, C. (2017). The persistent effect of temporary affirmative action. *American Economic Journal: Applied Economics*, 9(3):152–190.

OECD (2023). *Government at a Glance 2023*. OECD Publishing, Paris.

OECD (2024). Employment rate by sex and age group. OECD Data Explorer, Labour Force Statistics database.

Olivetti, C. and Petrongolo, B. (2016). The evolution of gender gaps in industrialized countries. *Annual Review of Economics*, 8:405–434.

Pitt, M. M., Rosenzweig, M. R., and Hassan, M. N. (2012). Human capital investment and the gender division of labor in a brawn-based economy. *American economic review*, 102(7):3531–3560.

Rosa, B. V. (2024). Subcontracting requirements and the cost of government procurement. *The RAND Journal of Economics*, 55(1):3–32.

Szerman, C. (2022). The labor market effects of disability hiring quotas. *Available at SSRN 4267622*.

A Matching and Sample

B Employment Outcome and the Female Share

For firm i, let F_{it} and M_{it} denote female and male employment, and let $N_{i,-1}$ denote baseline employment. Define

$$Y_{it}^{F} = \frac{F_{it}}{N_{i,-1}\overline{s}_{-1}^{F}}, \qquad Y_{it}^{M} = \frac{M_{it}}{N_{i,-1}\overline{s}_{-1}^{M}}.$$

Table A1: Matching Criteria

Criterion	Rule
Contracts	
Contract object	Same object: works, services, or supplies
Award procedure	Same procedure
Timing	Award within six months
Product classification	Same CPV at four, three, or two digits
Contract value	Log difference below 0.10 and absolute difference below €1,500
Firms	
Sector	Same macro-sector
Size	Log-employment difference below 0.30
Female share	Difference below 10 percentage points
Selection	Nearest standardized Euclidean distance, with replacement

Table A2: Matched and All Treated Firms

	Matched treated	All treated
Employment (median)	12	18
Female share of employment	0.111	0.109
Contract value (median, euros)	47,638	140,668
Construction share	0.734	0.709
Firm-contracts	685	4,112

Notes: The table compares matched treated firm-contracts with all eligible treated firm-contracts. Firm characteristics are measured in the baseline year.

For small changes in the female share $s = F/(F+M)$:

$$\Delta s \approx \overline{s}_{-1}^{F} \overline{s}_{-1}^{M} \left(\beta^{F} - \beta^{M} \right).$$

The female employment share therefore rises when the treatment effect on scaled female employment exceeds the effect on scaled male employment.